The formation of a RIFT VALLEY

▷ **DEEP AND HOT**
The lava lake of Nyiragongo volcano is sometimes 2,000 ft (600 m) deep. It connects to a magma reservoir in the crust beneath it, which has intruded there due to the rifting process.

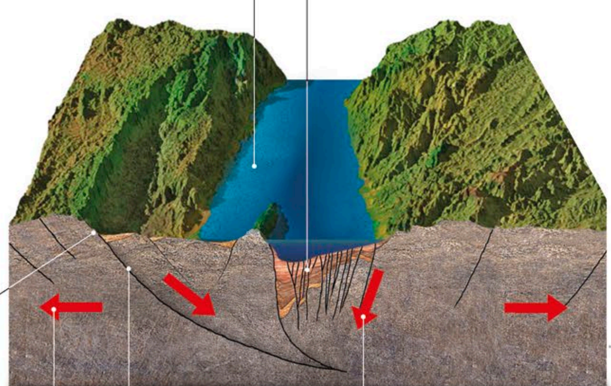
Continental rifting is a process in which part of one of Earth's tectonic plates, made up of the crust and upper mantle (together known as the lithosphere), becomes stretched and thinned to the point where fissures and faults appear. Blocks of crust collapse downward, leaving a series of depressions, called a rift valley, at the surface. The fissuring also allows magma (hot melted rock from the mantle) to intrude into the crust and create volcanic activity at the surface.

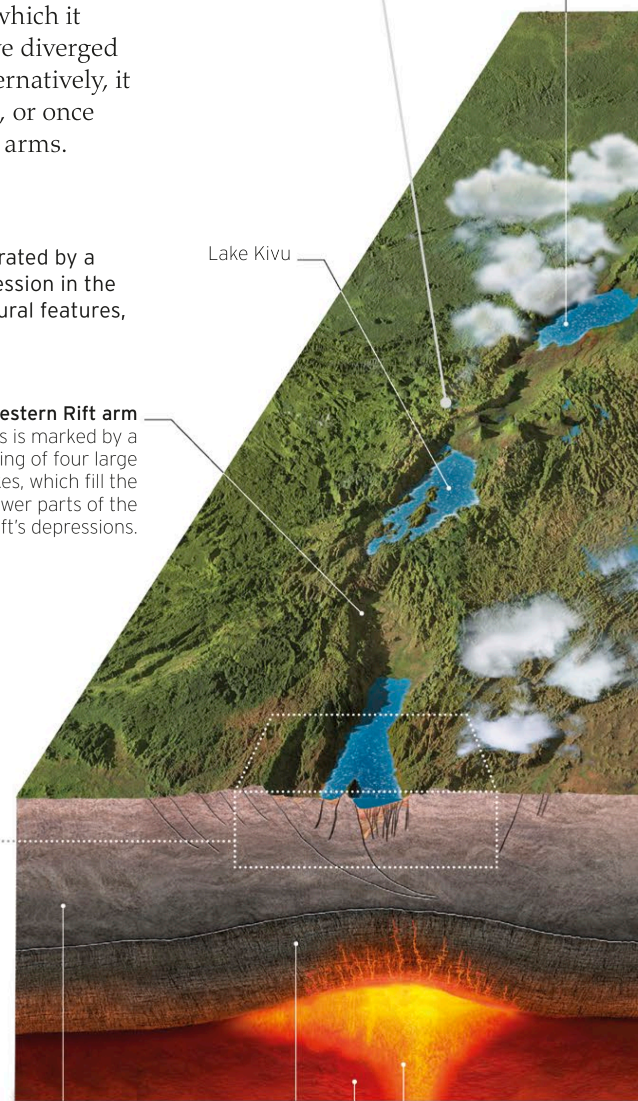
Rift valleys in East Africa

An excellent example of continental rifting can be seen in East Africa. The cause here is thought to be a mantle plume, or possibly plumes, under the region (see panel, below). The East African Rift system has two arms, east and west (see p.184), on either side of an area containing Lake Victoria. Rifting and volcanism started in a part of the eastern branch to the northeast of Lake Victoria around 35 million years ago, probably due to a mantle plume, then spread north and south. But as the rifting process encountered a stable core of lithosphere, around and to the south of Lake Victoria, which it could not go through, it is theorized to have diverged around it, leading to the two rift arms. Alternatively, it could be that separate mantle plumes exist, or once existed, under the eastern and western rift arms.

▶ **EAST AFRICAN RIFT SYSTEM**
In East Africa, two arms of a rift system are separated by a broad, stable plateau. Each arm is a curved depression in the landscape and the site of several remarkable natural features, many of them having a volcanic origin.

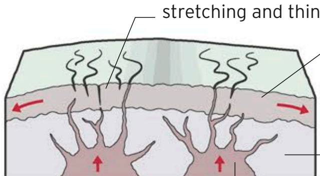
▷ **RIFT FAULTING**
In a rift, some blocks of crust drop down, along faults created by stretching forces. In the segment shown here, a long major fault has developed on one side. Along a whole rift, major faults may switch between sides, giving an overall sinuous appearance.





HOW THE EAST AFRICAN RIFT FORMED

The system of rift valleys in East Africa formed when a mantle plume, or possibly plumes, caused upward warping of the crust, then stretching and thinning as parts of the African Plate on either side began to move apart. Tensional faults appeared, and blocks of crust dropped down, producing depressions, which became partly filled by lakes, bordered by escarpments or mountains.



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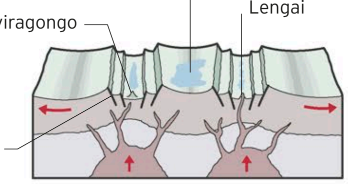

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